Plotting in Python

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Disclamer

- This is a presentation about **plotting libraries in Python**, without any visual examples.
- See the accompanying Jupyter Notebook for code snippets and rendered plots.
- This is not a deep dive into **how** to use each library, but rather **when** to use each library.

Lexicon

Imperative Specify *how* to draw step-by-step. Maximum control, verbose code.

Declarative Specify *what* to visualize; library handles execution. More concise, less control.

Grammar of Graphics Layered framework: Data $\rightarrow$ Aesthetics $\rightarrow$ Geometries.

Wrapper Library built atop a core library to simplify syntax and automate tasks.

Libraries Overview

Library	Role	Syntax	Output	Best For
		General Purpose		
Matplotlib	Core	Imperative	Static	Fine-tuning
Seaborn	Wrapper	Declarative	Static	Stats/EDA
Plotnine	Port	Grammar	Static	R Users
Plotly	Web	Declarative	HTML/JS	Dashboards
Altair	Web	Grammar (Vega)	HTML/JS	Linked Views
		Specialized & QoF		
UpSetPlot	Sets	Matrix Layout	Static	Intersections
Yellowbrick	ML	Scikit-API	Static	Model Diagnostics
YData Prof.	Auto	One-Liner	HTML Rep.	Instant Audit
Datashader	Big Data	Raster Pipeline	Image	50k+ Points

Publication Standards

Matplotlib

- Low-level control over every pixel.
- Uses an "Imperative" approach (you tell it *how* to draw).
- Verbose and complex, but indispensable for the final polish.

Perfect Context:
Use for granular control and final publication polishing.

Seaborn

- Built on top of Matplotlib
- It handles aggregation, confidence intervals, and color mapping automatically.
- The default for Cluster Maps and Violin plots.
- It transforms "Tidy Data" into complex biological visuals with single lines of code.

Perfect Context:
Use for rapid statistical summarization and cluster maps.

Plotnine

- A strict implementation of the "Grammar of Graphics" in Python.
- Almost exact clone of R's `ggplot2`.
- Allows team members fluent in R to write Python code without learning new syntax.
- Code Style: `(ggplot(data) + geom_point())`.

Perfect Context:
Use to bridge the gap between R and Python teams using the familiar "Grammar of Graphics" syntax.

Interactive & Web

Plotly

- Generates HTML/JS plots allowing zooming, panning, and hovering.
- Moves analysis from static images to dynamic exploration.
- Essential for dense datasets (e.g., PCA).
- Hover over an outlier dot to instantly see the Sample ID, Batch Number, and QC metrics.

Perfect Context:
Use for interactive exploration and outlier identification.

Altair

- Ideal for internal web tools.
- Supports "Linked Brushing"—selecting genes in a scatter plot automatically filters a connected bar chart.
- Link: https://altair-viz.github.io/gallery/index.html#example-gallery

Perfect Context:
Use for creating linked, interactive web visualizations.

Specialized Tools

UpSetPlot

- UpSet plots use a matrix layout to show intersections strictly.
- Comparing gene lists across many conditions.
- The rigorous way to visualize complex set intersections.

Perfect Context:
Use for deciphering complex overlaps when Venn diagrams fail.

Yellowbrick

- Visual diagnostic tools for Machine Learning.
- It visualizes the *model performance*, not just the data.
- Instantly generates ROC Curves, Confusion Matrices, and Class Balance charts.

Perfect Context:
Use for visual diagnostics of machine learning models.

Statannotations

- Removes the manual labor of drawing significance brackets and stars on plots.
- Automatically runs tests and draws the bracket with the correct p-value annotation.

Perfect Context:
Use for automating p-value annotations on comparison plots.

YData Profiling

- "Don't plot manually until you know what you have."
- Automates Exploratory Data Analysis (EDA).
- One line of code generates a full HTML audit showing missing values, correlations, and outliers.

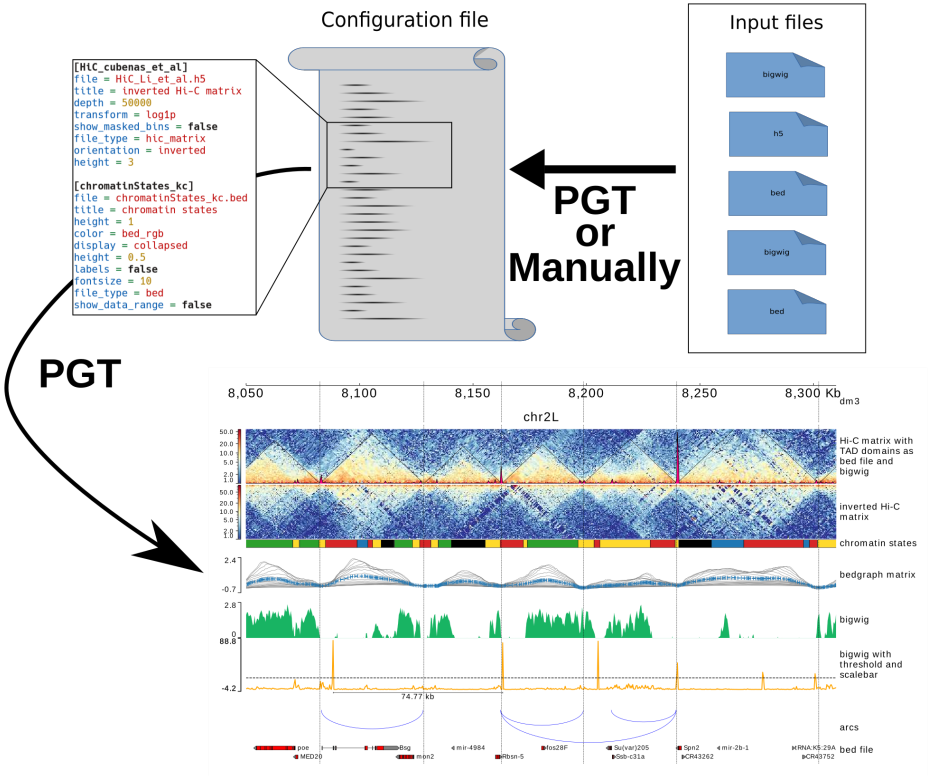
Perfect Context:
Use for an instant, comprehensive audit of new data.

Big Data & Genomics

Datashader

- Plotting 100,000+ cells (scRNA-seq) in Matplotlib creates a solid blob.
- Datashader rasterizes data into a heatmap *before* rendering.
- It aggregates millions of points into a single image in milliseconds.

Perfect Context:
Essential for Single-Cell UMAPs/t-SNE with more 50k cells where individual dots obscure density.

PyGenomeTracks

- A Python wrapper to generate publication-quality genomic tracks (ChIP-seq, ATAC-seq, RNA-seq).
- PyGenomeTracks is for *scripting* reproducible, vector-graphics quality track figures.